Reg. No.															

B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, NOVEMBER 2025

Fifth & Sixth Semester

21CSC303J - SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

(For the candidates admitted during the academic year 2021 - 2022 to 2024 - 2025)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks) **Marks CO BL**

Answer **all** Questions

1. Which of the following process models is known as a linear sequential model, where each phase must be completed before the next can begin? 1 1 1

   (A) Agile (B) Scrum

   (C) Waterfall (D) Spiral

2. The Agile process model is best suited for projects where: 1 1 2

   (A) All requirements are well-defined and unlikely to change.

   (B) The project team is geographically dispersed and works in isolation.

   (C) There is a need for flexibility and frequent adaptation to changing requirements.

   (D) Risk analysis is the most critical and dominant phase.

3. What is the primary characteristic of the "layered technology" approach in software engineering? 1 1 1

   (A) It focuses on a single, monolithic layer for all functions.

   (B) It builds on a foundation of quality, with each layer addressing different aspects of the process.

   (C) It prioritizes the end-user interface over internal code structure.

   (D) It is an alternative name for the Waterfall model.

4. The primary goal of the Scrum methodology is to: 1 1 2

   (A) Complete the entire project within a single, long phase.

   (B) Deliver a complete, bug-free product at the end of the project.

   (C) Maximize the value of the product by frequently delivering working increments.

   (D) Follow a rigid, pre-defined plan with no room for change.

5. A functional requirement primarily describes: 1 2 1

   (A) The quality attributes of the system, such as its performance and security.

   (B) The specific features and behaviors that the system must perform.

   (C) The constraints on the system's design and implementation.

   (D) The budget and timeline for the project.

6. The primary purpose of requirements elicitation is to: 1 2 1

(A) Validate the software requirements specification (SRS) document.
(B) Develop the user interface design for the system.
(C) Discover and gather the needs and constraints from various stakeholders.
(D) Estimate the cost and effort of the project using models.

7. Which of the following is an example of a non-functional requirement? 1 2 1

(A) The system must allow users to log in with a username and password
(B) The system must process a transaction in less than two seconds.
(C) The system must generate a report of all completed sales.
(D) The system must send an email notification to the user upon a successful order.

8. In the COCOMO model, the primary factor used to estimate effort is: 1 2 2

(A) The number of people on the development team.
(B) The number of months in the development schedule.
(C) The size of the software, typically measured in KLOC (Kilo Lines of Code).
(D) The number of identified functional requirements.

9. Which of the following is an example of an architectural pattern? 1 3 1

(A) A function that calculates a loan repayment.
(B) A database schema.
(C) A blueprint for a building.
(D) A Client-Server model.

10. What does the acronym UML stand for 1 3 1

(A) Unified Machine Language
(B) Universal Modeling Library
(C) Unified Modeling Language
(D) Universal Method Locator

11. Which of the following UML diagrams would be most useful for modeling the step-by-step interactions between a user and a system to achieve a specific goal, such as "logging in"? 1 3 1

(A) Class Diagram
(B) Sequence Diagram
(C) Use Case Diagram
(D) Component Diagram

12. A software architect chooses a Layered architectural style for a new e-commerce application. What is the primary benefit of this choice? 1 3 2

(A) It allows the system to be developed by a single person.
(B) It simplifies the user interface design.
(C) It promotes separation of concerns, making the system easier to maintain and extend.
(D) It guarantees the system will be free of bugs.

13. What is the primary purpose of a desk check in software construction? 1 4 1

(A) To execute the code with a debugger.
(B) To manually trace the execution of code on paper.
(C) To run a suite of automated tests.
(D) To review the user interface for usability.

14. Which of the following is a type of white-box testing? 1 4 1

(A) Boundary value analysis
(B) Equivalence partitioning
(C) Statement coverage
(D) Regression testing

15. A company enforces a coding standard that requires all variable names to be descriptive (e.g., `customerName` instead of `c`). What is the main benefit of this standard? 1 4 2

(A) It makes the code run faster.
(B) It reduces the number of security vulnerabilities.
(C) It improves code readability and maintainability.
(D) It prevents the need for code reviews.

16. A software team is using a test case to verify that a system's search function correctly handles both valid and invalid input . What type of testing strategy are they primarily using? 1 4 2

(A) Unit testing
(B) Black-box testing
(C) System testing
(D) Integration testing

17. What is the main characteristic of a proactive risk management strategy? 1 5 1

(A) It waits for risks to occur and then reacts to them.
(B) It identifies potential risks and plans to mitigate them before they happen
(C) It focuses on documenting risks after the project is complete.
(D) It ignores all potential risks to save time and resources.

18. Which of the following is a key activity in product release management? 1 5 1

(A) Designing the software's user interface.
(B) Writing the initial project requirements document.
(C) Building and packaging the final software for distribution.
(D) Estimating the project's total cost.

19. A software team discovers that a key library they depend on is no longer supported and has known security vulnerabilities. This is an example of which risk management activity? 1 5 2

(A) Risk projection.
(B) Risk identification.
(C) Risk mitigation.
(D) Risk monitoring.

20. The process of examining and altering an existing software system to re-create it in a new form without changing its behavior is known as: 1 5 1

(A) Maintenance.
(B) Reengineering.
(C) Testing.
(D) Deployment.

PART - B (5 × 8 = 40 Marks) **Marks CO BL**

Answer **all** Questions

21. a) Analyze and differentiate between the Scrum and Waterfall process models by comparing their approach to customer involvement, flexibility to change, and project delivery 8 1 4

(OR)

b) What is a process framework in software engineering? Describe its key components and explain its significance in software development 8 1 1

22. a) Compare and contrast the Basic and Intermediate COCOMO models, focusing on their primary purpose, the input variables they use for estimation, and the level of accuracy they provide 8 2 1

(OR)

b) You are assigned to a new project and asked to initiate the requirements engineering process. Describe the steps involved and analyze how a feasibility study supports the entire process. 8 2 4

23. a)	A small library is developing a new automated system to manage its book lending process. The system will allow a Librarian to add and remove new books from the database. A Member of the library can search for books, check their availability, and check out or return books. Create a high-level use case diagram and design a class Diagram that models the core structure of the system	8	3	4
	(OR)			
b)	Describe the following two architectural styles used in software design: (i) Layered Architecture (ii) Pipes and Filters Architecture	8	3	1
24. a)	Analyze and differentiate between Desk checks, Code walkthrough and Code review. For each method, explain its primary purpose, who is typically involved, and a key advantage it offers in the software development process.	8	4	4
	(OR)			
b)	Explain the differences between unit testing, integration testing, validation testing, and system testing. Analyze how these levels work together to ensure a reliable software product.	8	4	4
25. a)	Discuss the various risk identification techniques used in software project management. How does risk projection help in prioritizing risks during product release planning?	8	5	1
	(OR)			
b)	Define software maintenance and describe its major types. Why is maintenance considered a critical phase of the software lifecycle?	8	5	1

PART - C (1 × 15 = 15 Marks)	**Marks**	**CO**	**BL**
Answer **any 1** Question			
26. An organization wants to develop a Mobile Banking Application using Agile methodology. The project involves strict deadlines and compliance with security standards. Discuss how the roles of Product Owner, Scrum Master, and Development Team will contribute to: • Understanding user requirements, • Maintaining transparency through ceremonies (daily scrum, sprint review, retrospective), and • Delivering increments that meet business value.	15	1	4
27. Consider an Online Shopping Cart System that allows users to add items, update quantities, and checkout. Prepare five test cases for the "Add to Cart" and "Checkout" functionalities. Each test case must include test conditions, input data, expected result, and test type (unit/system/black-box/white-box). Justify why these test cases are critical for ensuring system quality.	15	4	5

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